

Section A

Answer **all** questions in the spaces provided on the question paper.

- 1** When copper is added to an acid, different products are obtained when different acids and conditions are used. For example, when copper reacts with dilute nitric acid, copper nitrate (V) and NO is produced. However if copper reacts with concentrated nitric acid, NO₂ is produced instead in addition to copper nitrate(V).

- (a)** Construct the half-equation for the reduction of nitrate to NO and hence write an balanced equation for this reaction.

[2]

- (b)** If 10 g of copper was reacted with dilute nitric acid, calculate

- (i)** volume of NO produced, measured at room temperature and pressure,

[2]

- (ii)** volume of 1.5 mol dm⁻³ nitric acid needed for complete reaction.

[2]

- 1 (c) Draw the dot and cross diagram for the following species and predict their shapes.



[2]



[2]



[2]

- (d) How will the bond angles in (c)(i) and (c)(ii) compare? Explain.

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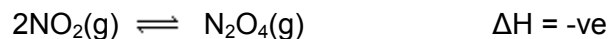
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[2]

- 1 (e) At room temperature and pressure NO_2 dimerises to form dinitrogen tetraoxide, N_2O_4 , as shown below:



- (i) NO behaves similarly. What will it form and why does this happen?

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[2]

- (ii) Write the expression for the equilibrium constant, K_c for the above equilibrium.

[1]

- (iii) At 298 K and 101 kPa, 1.00g of the mixture of these two gases at equilibrium occupying a volume of 0.317 dm^3 is found to show an average Mr of 77.3. The average Mr of the mixture can be computed using the following expression,

$$\text{Ave Mr} = \frac{[n_{\text{equilm}}(\text{NO}_2) \times \text{Mr}(\text{NO}_2)] + [n_{\text{equilm}}(\text{N}_2\text{O}_4) \times \text{Mr}(\text{N}_2\text{O}_4)]}{\text{Total number of moles at equilibrium}}$$

Fill in the table below and using the expression given above, solve for the value of y .

	NO_2	N_2O_4
Initial/ mol		0
Change/ mol	$-2y$	$+y$
Equilibrium/ mol		y

[1]

1 (e) (iv) Hence, determine the value of K_C .

[3]

(v) Deduce qualitatively the effect on the average M_r of this gaseous mixture when

the pressure is increased;

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[2]

the temperature is increased.

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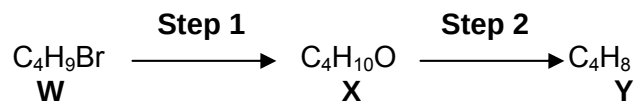
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[2]

[Total: 25]

- 2 Compound **W** reacts in two steps to form compound **Y**.



- (a) What are the reagents and conditions needed for steps 1 and 2?

Step 1.....

Step 2.....

[2]

- (b) (i) Compound **Y** is formed as a single isomer in the above reaction. If **Y** can be oxidised to form a carboxylic acid, suggest a structural formula for **W** and the corresponding **X**.

[2]

- (ii) State the reagent and condition needed to bring about the oxidation of **Y** and write a balanced equation for the oxidation of **Y**.

.....

[2]

- (iii) **W** can be converted into **Y** in a single step. What reagent and conditions are needed for this single step conversion?

.....

[1]

- 2 (c) (i) Another isomer of **W** also yields a single isomer of the product **Y** in the above sequence. What are the possible structures for **W**?

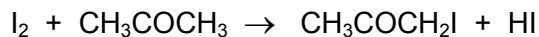
[2]

- (ii) If the product **Y** from (c)(i) is now oxidised, what will be the resulting organic product?

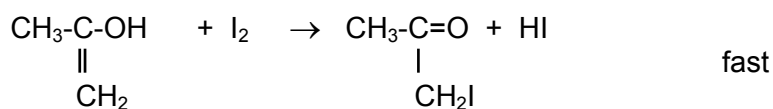
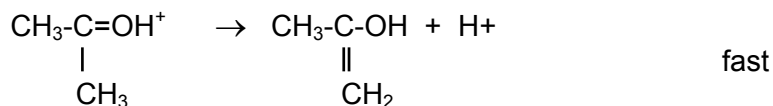
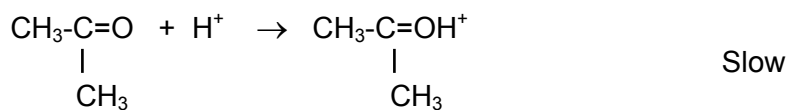
[1]

[Total: 10]

3 Iodine reacts with propanone as follows,



A possible mechanism for this reaction is



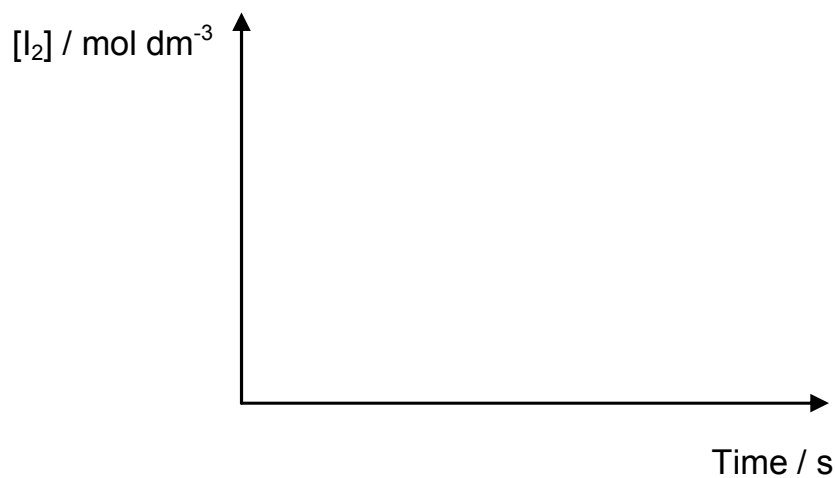
(a) Write a rate equation for this reaction based on the above mechanism.

..... [1]

(b) What evidence indicates that the reaction is acid catalysed?

.....
..... [1]

(c) On the axes below sketch the graphs you will obtain for the above reaction.



3 (c)



[2]

- (d) Iodine and propanone reacts differently if NaOH(aq) is present instead of $\text{H}^+(\text{aq})$. What are the resultant organic products in the new reaction?

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[1]

[Total: 5]

Section B

Answer **two** of the following three questions. Answer these questions on separate answer paper.

- 4 Sulfur is abundant in the Earth's crust. It occurs as elemental sulfur, as mineral sulfides and sulfates, and as organic sulfur compounds in oil and coal. However, sulfur is also a significant source of pollutant when it burns.

- (a) Describe the observation expected when
- the element reacts with oxygen
 - the resulting gaseous product bubbles into water containing universal indicator.

Write equations for the reactions described. [5]

- (b) Sulfur dioxide is also a good reductant. In the presence of water, sulfur dioxide is able to decolorize substances.

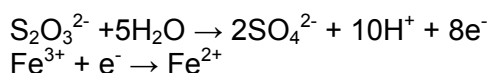
A solution of NH_4VO_3 in dilute H_2SO_4 can be reduced by SO_2 to give a deep blue solution. A 25.0 cm^3 of a solution containing 5.85 g dm^{-3} of NH_4VO_3 were reduced and then re-oxidised to the original state by 12.5 cm^3 of $0.0200 \text{ mol dm}^{-3}$ KMnO_4 . Deduce the oxidation state of the Vanadium in the deep blue solution. [5]

- (c) Thiosulfate occurs naturally and is produced by certain biochemical processes, but it can also be made from the reaction below.

- (i) The overall equation for the formation of thiosulfate ion from sulfite ion and sulfur in an alkaline solution is given below.
- $$\text{SO}_3^{2-}(\text{aq}) + \text{S}(\text{s}) \rightarrow \text{S}_2\text{O}_3^{2-}(\text{aq})$$

Write equations for the half-reactions that occur. [2]

- (ii) In an acidic medium, the thiosulfate is oxidised by $\text{Fe}^{3+}(\text{aq})$, which itself is reduced to $\text{Fe}^{2+}(\text{aq})$ according to the half-equations below.



In an experiment, 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ $\text{S}_2\text{O}_3^{2-}$ requires 20.0 cm^3 of the Fe^{3+} solution for complete reaction. What is the concentration of the Fe^{3+} solution? [4]

- (d) Use your knowledge of the chemistry of the elements in the Periodic Table to answer the following questions. Where possible, give reasons for your answers. [4]

- How does the atomic radius of sulfur compare with that of oxygen?
- How does the ionisation energy of sulfur compare with that of phosphorus?

[Total: 20]

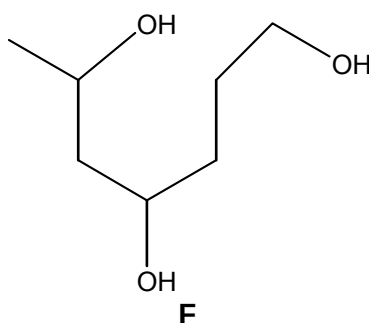
- 5 (a) Compound **A** ($C_{10}H_{16}$) contains a 6-membered carbon cyclic ring.

On hydrogenation, in the presence of nickel catalyst, compound **B** ($C_{10}H_{20}$) is formed. On hydration, compound **A** forms compound **C**, a diol, which cannot undergo further oxidation by hot acidified potassium dichromate(VI).

Compound **A** can be oxidised by hot acidified potassium manganate(VII) to compounds **D** and **E**.

Compound **D** contains 62.1 % C, 10.0 % H and 27.6% O. Compound **D** produces an orange precipitate on undergoing condensation with 2,4-dinitrophenylhydrazine.

Compound **E** can be reduced to give the compound **F** below.



Suggest, with reasons, the structures of compounds **A**, **B**, **C**, **D**, and **E**.

[10]

- (b) Compound **F** can be used as a fuel as well as a solvent.

An experiment was carried out as follows to determine the standard enthalpy change of combustion of compound **F**, contained in a gas cylinder.

A large beaker of water was heated by burning compound **F**. The temperature rise was recorded. The cylinder was weighed before and after the experiment to determine the mass of compound **F** used. The following results were obtained.

Mass of F used/ g	2.5
Mass of water heated/ g	500
Initial temperature / °C	30
Final temperature/ °C	90

- (i) Define the term enthalpy change of combustion. [1]
- (ii) Write a balanced equation for the combustion of compound **F** including state symbols. [1]
- (iii) The use of the *Data Booklet* is relevant to this question.

Use the data given to calculate the enthalpy change of combustion of compound **F**.

[3]

- 5 (b) (iv) Use the bond energies given in the *Data Booklet* to calculate another value of the enthalpy change of combustion of compound **F**. [3]
- (v) Suggest a reason for the discrepancy between the calculated values in (b)(iii) and (b)(iv). [1]
- (vi) Compound **F** is miscible with water. Draw a diagram, including dipoles, to show the hydrogen bonding between a molecule of **F** and a molecule of water. [1]

[Total: 20]

- 6 (a) Compound **G** is a sweet-smelling cyclic compound of molecular formula $C_6H_8O_4$.

Compound **G** reacts with dilute sulfuric acid on heating to yield only one product, **H**. Compound **H** reacts with sodium carbonate to give a colourless and odourless gas which gives white precipitate with limewater and with PCl_5 to form dense white fumes.

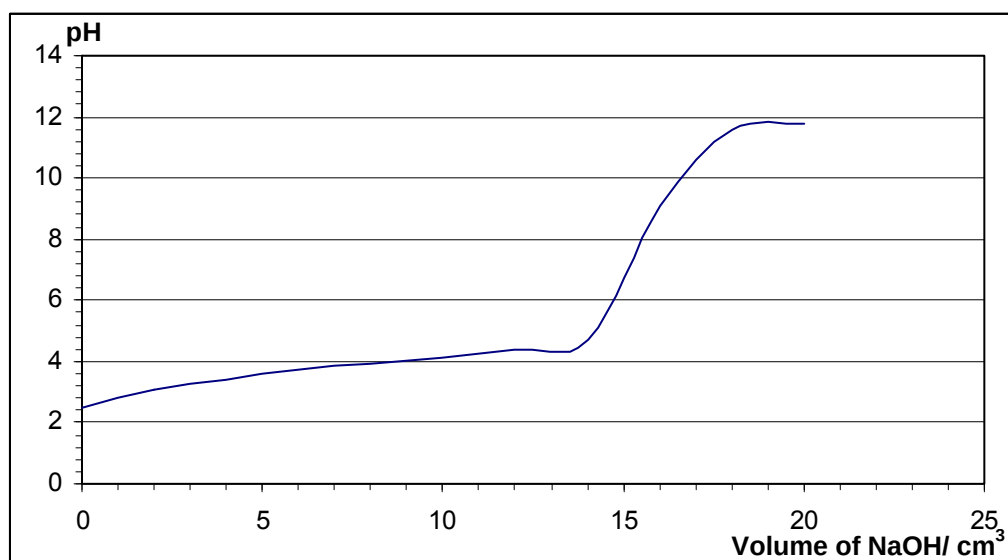
Upon reduction, compound **H** forms compound **K**, of molecular formula $C_3H_8O_2$. Compound **K** produces a yellow precipitate on reaction with alkaline aqueous sodium hydroxide. 1 mole of compound **K** reacts with 2 moles of sodium to yield 1 mole of hydrogen gas.

Compound **G** can also be reduced by $LiAlH_4$ to compound **K**.

[9]

Suggest, with reasons, the structures of compounds **G**, **H** and **K**.

- (b) When 10.0 cm^3 of a solution of **H** was titrated against 0.050 mol dm^{-3} of sodium hydroxide, the following graph is obtained.



- (i) For this titration, suggest a suitable indicator to determine the end-point of the reaction. [2]
- (ii) Calculate the concentration of **H** in the solution in mol dm^{-3} . [3]
- (iii) A buffer mixture is obtained when **H** is titrated against sodium hydroxide. Explain what is meant by the term buffer solution. [2]
- (iv) State the volume, with reason, at which maximum buffer capacity occurs when sodium hydroxide added. [2]
- (v) Deduce the K_a value for **H**, giving your reasoning and the units. [2]

[Total: 20]